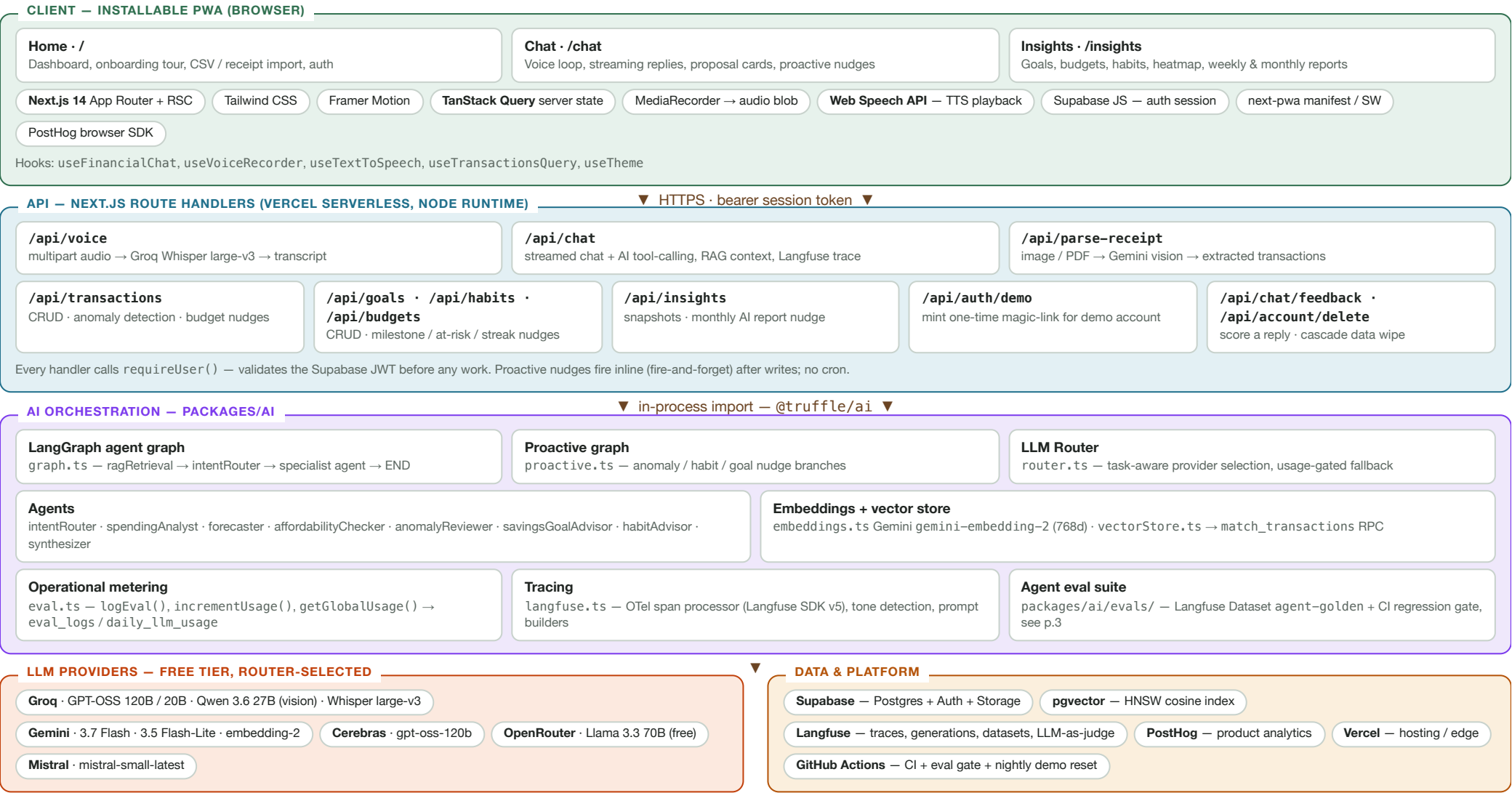


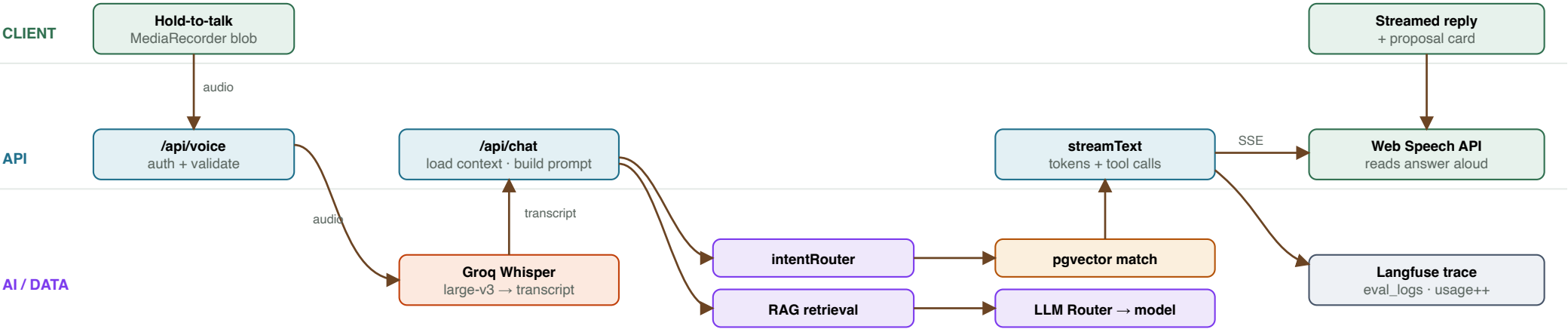
System overview

A user speaks or types; Whisper transcribes; a LangGraph agent graph classifies intent and reasons over the user's real transaction history; a task-aware multi-provider LLM router picks the best available free-tier model; the answer streams back and is read aloud. Every LLM call is traced to Langfuse and metered in Postgres, and every push to main re-runs the agent suite through a Langfuse-backed CI quality gate.



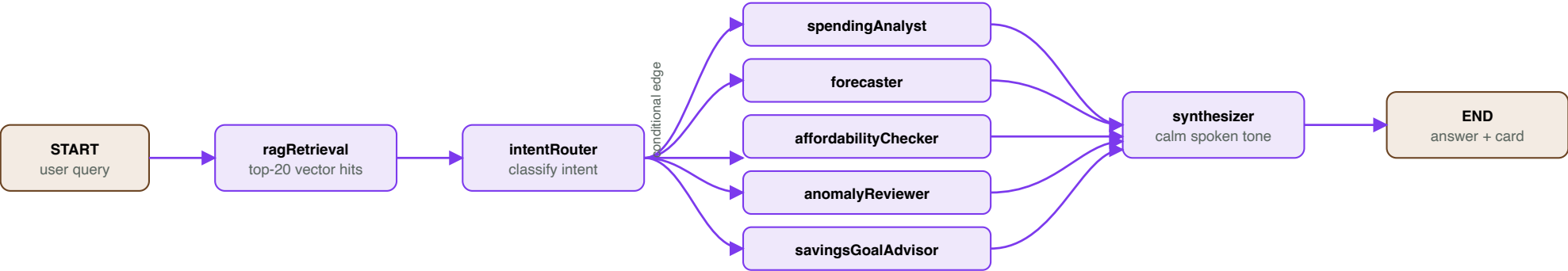
Client API routes AI orchestration LLM providers Data & platform

End-to-end voice request



The client sends recent context with every turn. /api/chat loads the last 50 transactions, snapshots, anomalies, goals, habits and budgets from Supabase in parallel, runs intent classification + semantic retrieval, builds a system prompt, then streams the reply. Tool calls (proposeGoal, proposeTransaction, proposeHabit) render confirmation cards the user taps to commit.

LangGraph agent graph — packages/ai/src/graph.ts



routeAfterIntent() maps the classified intent to exactly one specialist node. Each node calls routedGenerateText(taskType, options), which selects a provider, runs the call, logs to eval_logs, and increments the daily usage counter. The proactive.ts graph reuses the same annotation and agents for unsolicited nudges (anomaly / habit-streak / goal milestone), written straight to chat_messages with a unique nudge_key. **Note:** this graph itself has no live caller yet — spendingAnalyst, forecaster and affordabilityChecker only run today through the CI eval suite (below) and are exercised directly, not via buildTruffleGraph().

Task-aware LLM router — router.ts

On every call the router reads today's global per-provider request counts from Postgres and walks the priority list for that task type, skipping any provider already at its daily budget. First one under budget wins; streaming callers get the full ordered candidate list and fall through on per-minute rate limits.

TASK TYPE		PRIORITY ORDER (FIRST AVAILABLE WINS)
fast-chat		Groq → Cerebras → OpenRouter → Mistral
reasoning		Gemini → Groq → Cerebras → OpenRouter
tool-calling		Groq → Gemini → Cerebras
vision		Gemini → Groq

PROVIDER	DAILY CAP	MODELS
Groq	900	gpt-oss-120b / 20b · qwen3.6-27b · whisper-large-v3
Gemini	1 300	gemini-3.7-flash · 3.5-flash-lite · embedding-2
Cerebras	1 500	gpt-oss-120b
OpenRouter	250	llama-3.3-70b-instruct:free
Mistral	600	mistral-small-latest

Counters are incremented via the race-safe increment_llm_usage Postgres RPC. Caps sit ~10% under real free-tier limits.

Agent eval suite & CI gate — packages/ai/evals/

Rebuilt 2026-09 on Langfuse Datasets + Experiments (replaced an assertion-free golden script). Dataset agent-golden (27 items: 15 router-intent cases + 2 per LLM-calling agent) is source-controlled in dataset-items.ts and synced via pnpm eval:sync-dataset.



EVALUATOR	CHECKS
intentExactMatch	deterministic — routeIntent() output equals the expected QueryIntent
numericFaithfulness	deterministic — response cites the figure computeMonthProjection() actually computed (± €1), not the model's own mental math
responseQualityJudge	LLM-as-judge — mirrors the live response-quality Langfuse evaluator, 1–5 rubric

CI gate — langfuse-eval.yml via langfuse/experiment-action, on every push to main (no PR flow here); fails the job if avg_intent_accuracy < 0.6, avg_response_quality < 3.0, or avg_numeric_faithfulness < 0.75. Concurrency capped at 2 for Gemini's free-tier quota. Confirmed green on main, 2026-09-19.

Separately: the always-on production **response-quality** evaluator scores live streamText / adviseHabit / adviseSavingsGoals / reviewAnomalies generations async in Langfuse; categorize-eval.ts (Weco-tunable) scores CATEGORY_GUIDANCE against a transactions.csv sample.

Persistence — Supabase Postgres

TABLE	HOLDS
transactions	ledger rows + embedding vector(768)
anomalies	detected unusual charges (type, severity)
monthly_snapshots	per-month income / expense / category rollup (JSON)
chat_messages	conversation + proactive nudges (is_proactive, nudge_key, read_at)
savings_goals	target, saved, deadline, emoji
savings_habits	recurring weekly / monthly saving rule
habit_contributions	per-period contribution log (streaks)
monthly_budgets	per-category spending limit
daily_llm_usage	global per-provider request counter (per day)
eval_logs	operational log per LLM call — provider, task, latency, tokens (quality/intent scoring moved to Langfuse 2026-09, dead columns dropped)

RPC / EXTENSION	PURPOSE
pgvector + HNSW	cosine-distance ANN index on transactions.embedding
match_transactions	semantic retrieval for RAG context
increment_llm_usage	atomic, race-safe usage increment

Row-level security scopes every table to auth.uid(). Account deletion cascades via FK constraints. Auth = Supabase magic-link (passwordless); the demo account uses a server-minted one-time token.

Monorepo & delivery

PACKAGE	ROLE
apps/web	Next.js 14 PWA — pages, API routes, components, hooks
apps/landing	marketing site (Next.js, port 3001)
packages/ai	router, LangGraph, agents, prompts, embeddings, eval suite, Langfuse
packages/db	Supabase client + schema types + SQL migrations
packages/types	shared TS interfaces (Transaction, MonthlySnapshot, ...)

- **Build** — pnpm workspaces + Turborepo; **deploy** — Vercel.
- **CI** (ci.yml) — type-check, lint, Vitest units, Playwright e2e on every push.
- **Eval gate** (langfuse-eval.yml) — agent-golden regression check on every push to main.
- **Nightly** (demo-reset.yml) — GitHub Action re-seeds the demo account from transactions.csv.